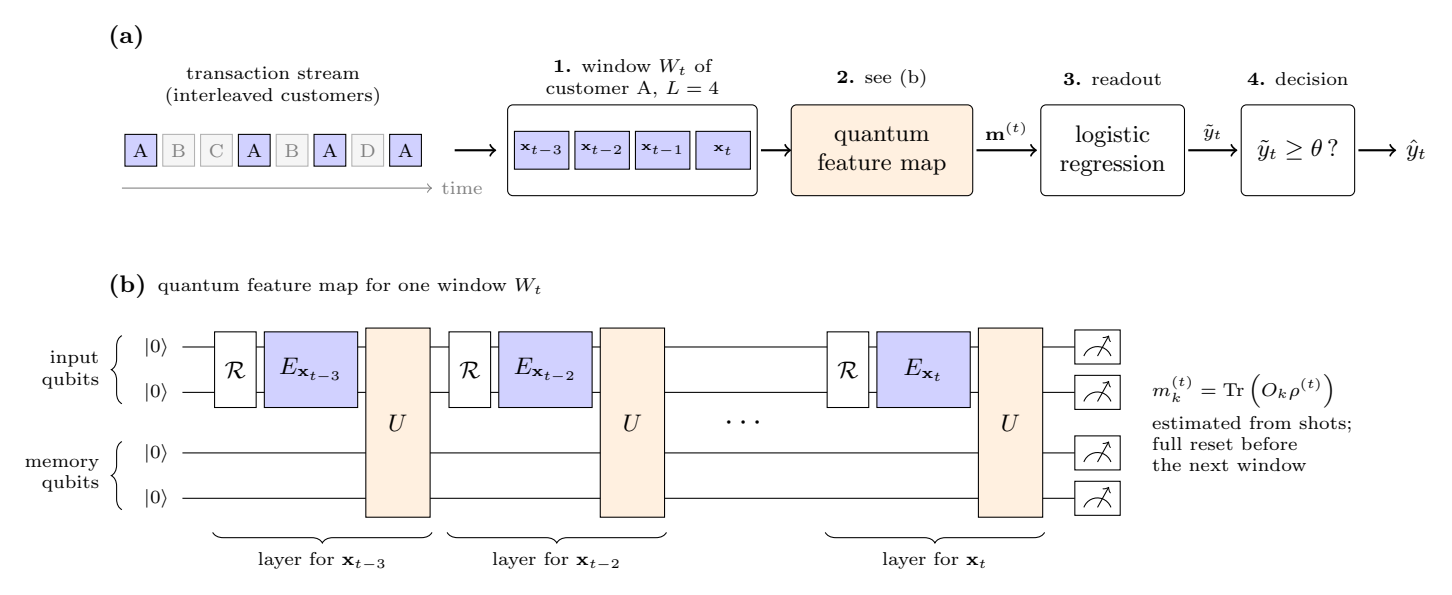

(a)
transaction stream
(interleaved customers)
A B C A B A D A
time
1. window $W_t$ of
customer A, $L = 4$
$\mathbf{x}_{t-3}$ $\mathbf{x}_{t-2}$ $\mathbf{x}_{t-1}$ $\mathbf{x}_t$
2. see (b)
quantum
feature map
$\mathbf{m}^{(t)}$
3. readout
logistic
regression
$\tilde{y}_t$
4. decision
$\tilde{y}_t \geq \theta\,?$
$\hat{y}_t$
(b) quantum feature map for one window $W_t$
input
qubits
memory
qubits
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$\mathcal{R}$
$E_{\mathbf{x}_{t-3}}$
$U$
$\mathcal{R}$
$E_{\mathbf{x}_{t-2}}$
$U$
$\cdots$
$\mathcal{R}$
$E_{\mathbf{x}_t}$
$U$
layer for $\mathbf{x}_{t-3}$
layer for $\mathbf{x}_{t-2}$
layer for $\mathbf{x}_t$
$m_k^{(t)} = \mathrm{Tr}\left(O_k \rho^{(t)}\right)$
estimated from shots;
full reset before
the next window